SF

KPK BOARD TEXTBOOK NOTES - MATHEMATICS CLASS 9

Unit 17: Practical Geometry - Triangles

Complete, easy English notes based on *A Textbook of Mathematics for Grade IX*, Khyber Pakhtunkhwa Textbook Board (KPTB), Peshawar - constructing a triangle in the three given cases, drawing angle bisectors, altitudes, perpendicular bisectors and medians and verifying their concurrency, and constructing figures equal in area, with the book's own examples, steps of construction and exam practice.

Notes by **Shah Fahad**
Primary School Teacher (PST), ESED Khyber Pakhtunkhwa - Buner, Pakistan
Contact: 03409090945 | November 2026

> **How to use these notes:** This is a drawing unit. Marks are given for a neat, accurate figure *and* for written steps of construction, so practise both together: draw with a sharp pencil, keep every compass arc visible, label every point, and write the steps in order as you draw. Never rub out the arcs - they are the proof that your construction is correct.

What this unit covers

1. Tools of construction and the six elements of a triangle
2. Case 1 - two sides and the included angle (SAS)
3. Case 2 - one side and two angles (ASA)
4. Case 3 - two sides and an angle opposite one of them (SSA): one, two, or no triangle
5. Concurrent lines - angle bisectors and the incentre
6. Altitudes and perpendicular bisectors - orthocentre and circumcentre
7. Medians and the centroid, and the four centres compared
8. Figures with equal areas - quadrilateral to triangle, triangle to rectangle
9. Square equal to a rectangle, and a triangle of equal area on a given base
10. Summary, common mistakes, MCQs and construction practice with answers

Source book: KPTB Mathematics for Grade IX, Unit 17 (book pages 298-312). The notes follow the book's order, examples and exercises. The scan used for verification carries no printed copyright page, so no printing year is claimed here - the 17-unit structure is the current circulating KPTB edition.

1. Tools and the Six Elements of a Triangle

Practical geometry is the link between theory and practice - engineers, architects and builders use it before they build. In this unit a **construction** means a drawing made with only two tools: a **straight edge (ruler)** for drawing lines, and a **pair of compasses** for drawing arcs and copying lengths. A protractor is used in class to check angles, but the constructions themselves are compass constructions.

The six elements

A triangle has **six elements: three sides and three angles**. For triangle ABC they are the sides AB, BC and CA, and the angles at A, at B and at C. To construct a triangle we do not need all six - three well-chosen elements are enough, and Sections 2 to 4 are the three combinations the book gives.

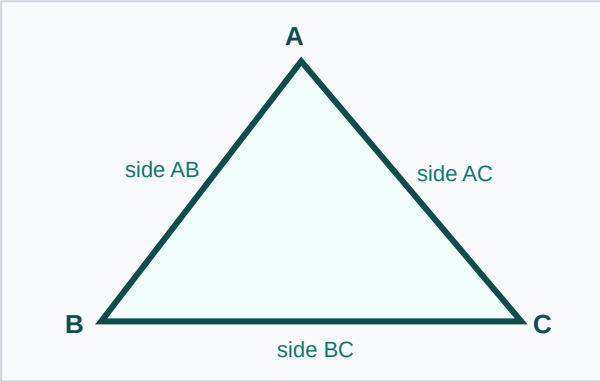


Figure 1. The six elements of triangle ABC: sides AB, BC, CA and angles A, B, C.

Four basic constructions you must already know

- **Bisect a line segment:** arcs of equal radius (more than half the segment) from both ends, above and below; join the two crossing points.
- **Bisect an angle:** put the compass point on the vertex - this is always the first step - draw an arc cutting both arms, then equal arcs from those two points; join the vertex to their crossing point.
- **Draw a perpendicular** from a point to a line, and **copy a given angle** at a new point, both by the standard equal-arc method.
- **Construct standard angles** with compasses: 90 degrees and 60 degrees first, then bisect them to get 45 degrees and 30 degrees, and combine them for 75 degrees, 105 degrees and 120 degrees.

> **Not every three elements work:** three angles alone do *not* fix one triangle - they fix only its shape, so infinitely many triangles of different sizes fit the same three angles. At least one side must always be given, because a side fixes the size.

2. Case 1 - Two Sides and the Included Angle (SAS)

The **included angle** is the angle between the two given sides - the angle at their common vertex. When two sides and their included angle are given, exactly **one** triangle can be constructed. This is the same data as the SAS congruence rule of Unit 10.

Example 1 (book). Construct triangle PQR given that PQ = 4 cm, PR = 5 cm and angle P = 120 degrees.

Steps of construction:

1. Draw PQ of measure 4 cm.
2. At P, construct an angle of 120 degrees with the help of a compass.
3. With P as centre, draw an arc of radius 5 cm cutting the new arm at R, so PR = 5 cm.
4. Join R to Q. PQR is the required triangle.

Check your drawing afterwards: measure PR (5 cm), angle P (120 degrees) and the third side QR. If any one disagrees, the error is in the angle construction or in the compass opening - find it before going on.

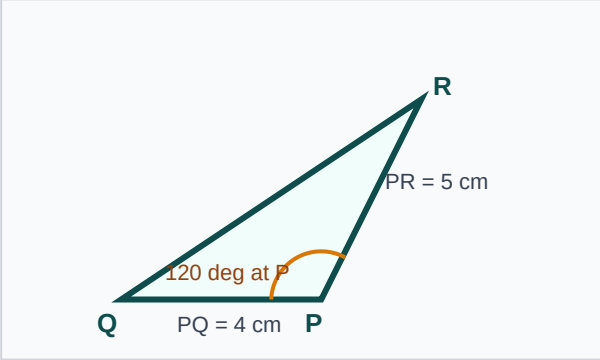


Figure 2. Case 1: the included angle at P sits between sides PQ and PR (figure not to scale).

The general recipe for Case 1

1. Draw one given side as the base.
2. At their common vertex, construct the included angle.
3. Cut the new arm at the length of the second side with a compass arc.
4. Join the free ends. Write the steps, in this order, beside the figure.

> **Exam tip:** If the given angle is *not* between the two given sides, it is not Case 1 - it is Case 3 (Section 4), where the answer may be two triangles or none. Always ask first: is the angle included?

3. Case 2 - One Side and Two Angles (ASA)

When one side and two angles are given, draw the side first and build one angle at each end; where the two new arms meet is the third vertex. Again exactly **one** triangle is possible, matching the ASA congruence rule.

Example 2 (book). Construct triangle PQR such that PQ = 5.4 cm, angle PQR = 45 degrees and angle RPQ = 60 degrees.

Steps of construction:

1. Draw PQ measuring 5.4 cm.
2. At P, draw an angle of 60 degrees, and at Q, draw an angle of 45 degrees.
3. Let the two arms meet at R.
4. PQR is the required triangle.

A useful check: the third angle must be 180 - 60 - 45 = **75 degrees**. Measure angle R in your figure - if it is not about 75 degrees, one of the two drawn angles is wrong.

When the two angles are not both on the given side

Sometimes the given angles sit at vertices that are not the two ends of the given side - for example, side AB with angles A and C given. The method does not change; only one extra step is needed first:

1. Find the third angle from the angle sum: angle B = 180 degrees - (angle A + angle C).
2. Now the side has a known angle at both ends, and the construction is exactly Example 2.

Example 3. Construct triangle ABC with AB = 6 cm, angle A = 70 degrees and angle C = 50 degrees (the book's median example begins this way).

First angle B = 180 - (70 + 50) = 60 degrees. Draw AB = 6 cm, construct 70 degrees at A and 60 degrees at B, and let the arms meet at C. The triangle is fixed, and angle C measures 50 degrees as required.

Impossible data: If the two given angles add to 180 degrees or more, the two arms never meet on the same side and **no triangle can be constructed**. Check the sum before you start drawing - it takes five seconds and saves a wasted page.

4. Case 3 - Two Sides and an Opposite Angle (SSA)

Here the given angle is **opposite** one of the given sides, not included between them. This is the **ambiguous case**: the compass arc that fixes the third vertex may meet the angle's arm once, twice, or not at all, so there can be **one triangle, two triangles, or no triangle**. The book shows all three outcomes.

Book example	Given data	What the arc does	Result
Example 3	BC = 3 cm, AB = 6 cm, angle A = 30 deg	Arc from B, radius 3 cm, touches the arm at only one point C	One triangle ABC
Example 4	KL = 6.5 cm, KM = 5.5 cm, angle L = 45 deg	Arc from K, radius 5.5 cm, cuts the arm at two points M and M'	Two triangles KLM and KLM'
Example 5	ZX = 4 cm, YZ = 6.5 cm, angle Y = 60 deg	Arc from Z, radius 4 cm, never reaches the arm	No triangle possible

Steps for Example 4 - the two-triangle case

1. Draw KL = 6.5 cm.
2. At L, construct angle KLX = 45 degrees.
3. With K as centre, draw an arc of radius 5.5 cm to intersect ray LX at M and M'.
4. Join K to M and to M'. Both KLM and KLM' satisfy the given data, so both are required triangles.

> **How to predict the outcome:** Compare the side opposite the given angle with the perpendicular distance from the other vertex to the arm. If the arc's radius is *shorter* than that distance, it cannot reach - no triangle. If it *equals* it, it touches once - one (right-angled) triangle. If it is *longer* but still shorter than the adjacent side, it cuts twice - two triangles. This is why SSA is **not** a congruence rule: the same data can build two different triangles.

> **Exam trap:** In a Case 3 question, never stop after the first intersection. Look for the second crossing point, and if the arc does not meet the arm at all, write clearly: *no triangle can be constructed with the given data* - that statement, with the attempted figure, is the full answer.

5. Concurrent Lines and Angle Bisectors

Two lines in a plane are either parallel or intersecting, and two intersecting lines meet at exactly one point. If a third line also passes through that same point, the lines are called **concurrent**, and the common point is the **point of concurrency**. The rest of this unit verifies, by drawing, that four famous sets of three lines in a triangle are each concurrent.

Angle bisector: a line which divides a given angle into two angles of equal measure. The three angle bisectors of a triangle are concurrent. Their point of concurrency is called the **incentre**, and it is equidistant from the three **sides** of the triangle.

Example 6 (book). Construct triangle ABC with AB = 4.6 cm, BC = 5 cm and CA = 5.1 cm. Draw the angle bisectors and verify that they are concurrent.

Steps of construction:

1. Draw BC measuring 5 cm.
2. With B and C as centres and radii 4.6 cm and 5.1 cm respectively, draw arcs intersecting at A.
3. Join AB and AC. Triangle ABC is constructed (this is the three-sides method: two arcs fix the third vertex).
4. Bisect angles A, B and C by the standard method: compass point on the vertex, an arc cutting both arms, then equal arcs from those cut points.
5. The three bisectors pass through the same point - call it I. The angle bisectors are concurrent at the incentre I.

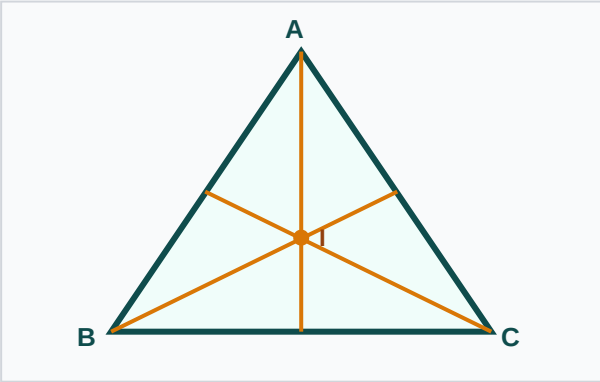


Figure 3. The three angle bisectors meet at one point, the incentre I.

Because I is equidistant from the sides, it is the centre of the circle that touches all three sides from inside (the incircle) - a fact the book's Review Exercise tests directly.

6. Altitudes and Perpendicular Bisectors

Altitudes - the orthocentre

An **altitude** is the perpendicular drawn from a vertex of a triangle to its opposite side (produced if necessary). In the book's Example 7, triangle ABC has AB = 5.6 cm, BC = 6 cm and CA = 5 cm; perpendiculars are drawn from A to BC, from B to CA and from C to AB, and the three altitudes AD, BE and CF all pass through the same point O. That point is called the **orthocentre**.

> **Obtuse triangles:** In an obtuse-angled triangle the orthocentre falls *outside* the triangle, and two altitudes must be drawn to the sides produced. In a right-angled triangle it falls exactly on the right-angle vertex. Draw the first two altitudes accurately - the third must pass through their crossing point, which is also your accuracy check.

Perpendicular bisectors - the circumcentre

A **perpendicular bisector** (or right bisector) is a line which is perpendicular to a line segment and divides it into two equal parts. To draw one: from both ends of the segment, with equal radius more than half the segment, draw arcs above and below, and join their two crossing points.

Example 8 (book). Construct triangle KLM with KL = 5.8 cm, LM = 6 cm and angle L = 60 degrees (a Case 1 construction). Draw its perpendicular bisectors and verify their concurrency.

1. Construct triangle KLM from the given data.
2. Draw the right bisectors of KL, LM and MK.
3. All three pass through one point - the **circumcentre** - so they are concurrent.

The circumcentre is equidistant from the three **vertices**, so it is the centre of the circle passing through A, B and C (the circumcircle). Notice the contrast with Section 5: incentre - equidistant from the *sides*; circumcentre - equidistant from the *vertices*. Examiners swap these two on purpose.

7. Medians and the Four Centres Compared

A **median** is the line segment joining the midpoint of one side of a triangle to its opposite vertex. To draw the medians, first find the midpoints of the three sides (bisect each side, as in Section 1), then join each midpoint to the opposite vertex.

Example 9 (book). Construct triangle ABC with AB = 6 cm, angle A = 70 degrees and angle C = 50 degrees. Draw its medians and verify their concurrency.

1. First find angle B = 180 - (70 + 50) = 60 degrees, then construct the triangle by Case 2.
2. Find the midpoints D, E and F of BC, CA and AB respectively.
3. Draw the medians AD, BE and CF.
4. All three medians pass through one point G. The medians of a triangle are concurrent at the **centroid** G.

The centroid divides each median in the ratio **2 : 1**, the longer part being towards the vertex: for example AG : GD = 2 : 1. The centroid is also the balancing point (centre of gravity) of a triangular plate.

The four points of concurrency - learn this table

Lines drawn	Point	Special property
Angle bisectors	Incentre (I)	Equidistant from the three sides; centre of the incircle
Perpendicular bisectors of the sides	Circumcentre	Equidistant from the three vertices; centre of the circumcircle
Altitudes	Orthocentre (O)	Inside for acute, on the vertex for right-angled, outside for obtuse triangles
Medians	Centroid (G)	Divides each median in the ratio 2 : 1 from the vertex

Special case (Review Exercise): In an **equilateral** triangle all four points are the *same* point, because every median is also an altitude, an angle bisector and a perpendicular bisector. That is the answer to the book's question about a triangle whose orthocentre, centroid and circumcentre coincide.

8. Figures with Equal Areas - I

Every construction in this part rests on one result from Unit 16: **triangles (and parallelograms) on the same base and between the same parallel lines are equal in area**, because area = (1/2) x base x height and both the base and the height are the same. Moving a vertex along a line parallel to the base changes the shape but never the area.

8.1 Triangle equal in area to a given quadrilateral

Given: quadrilateral ABCD. **Required:** a triangle equal in area to ABCD.

1. Join C to A (a diagonal of the quadrilateral).
2. Through D, draw DP parallel to CA, meeting BA produced at P.
3. Join C to P.
4. Triangles APC and ADC stand on the same base AC and between the same parallels AC and PD, so their areas are equal. Adding triangle ABC to both: area of triangle PBC = area of quadrilateral ABCD. Triangle PBC is the required triangle.

8.2 Rectangle equal in area to a given triangle

Given: triangle ABC. **Required:** a rectangle equal in area to ABC. The idea: a rectangle on *half* the base with the same height has the same area as the triangle, because (1/2) x base x height = (base/2) x height.

1. Let ABC be the given triangle.
2. Bisect AB at D.
3. At D, construct angle ADE = 90 degrees (the perpendicular at D).
4. Through C, draw a line parallel to BA, meeting DE produced at G.
5. Through A, draw a line parallel to DG, meeting the line through C at H.
6. ADGH is the required rectangle, equal in area to triangle ABC.

> **Written-answer pattern:** For every equal-area question, write Given, Required, Steps of Construction, and one or two lines of justification naming the same base and the same parallels. The justification lines are where the theory marks are.

9. Figures with Equal Areas - II

9.1 Square equal in area to a given rectangle

Given: rectangle ABCD. **Required:** a square of the same area. If the rectangle is l by w, the square's side s must satisfy s^2 = l x w, so s is the **geometric mean** of l and w - and a semicircle is the compass construction that produces it.

1. Produce AD to E such that DE = CD (the two adjacent sides now lie on one straight line AE).
2. Bisect AE at O.
3. With centre O and radius OA, draw a semicircle.
4. Produce CD to meet the semicircle at M.
5. On DM as a side, complete the square DFLM. This square is equal in area to rectangle ABCD, because DM^2 = AD x DE = AD x CD.

Check with numbers (Exercise 17.3, Q5). A rectangle has adjacent sides 4.5 cm and 2.2 cm. Find the side of the equal square.

Area of rectangle = 4.5 x 2.2 = 9.9 cm^2. Side of square = sqrt(9.9), which is about **3.15 cm**, and 3.15^2 is about 9.9 cm^2 - the areas agree, as the exercise asks you to verify by measuring your constructed square.

9.2 Triangle of equal area on a base of given length

Given: triangle ABC with base BC, and a required new base length x. **Required:** a triangle of area equal to ABC standing on a base of length x.

1. Let ABC be the given triangle.
2. Through A, draw a line AD parallel to BC. Any triangle with vertex on this line and base on BC has the same height, hence the same area.
3. With B as centre, draw an arc of radius x cutting line AD at M.
4. Join M to B and to C. Triangle BCM is the required triangle: its chosen base BM has length x, and its area equals the area of triangle ABC.

Chain of conversions: The four constructions connect: quadrilateral -> equal triangle (8.1) -> equal rectangle (8.2) -> equal square (9.1), and any triangle can be rebuilt on a given base (9.2). Exercise 17.3 combines them, for example turning a given triangle into an equal square by passing through the rectangle.

10. Summary and Common Mistakes

Construction	Key step to remember
Case 1 (SAS)	Draw a side, build the included angle at the common vertex, cut the arm with an arc of the second side
Case 2 (ASA)	Draw the side, build an angle at each end; if an angle is not on the side, find the third angle first (180 deg rule)
Case 3 (SSA)	Draw the side and angle, then count the arc's intersections: 1 = one triangle, 2 = two triangles, 0 = none
Angle bisectors	Concurrent at the incentre - equidistant from the sides
Altitudes	Concurrent at the orthocentre - perpendiculars from vertices to opposite sides
Perpendicular bisectors	Concurrent at the circumcentre - equidistant from the vertices
Medians	Concurrent at the centroid - divides each median 2 : 1
Equal areas	Same base + same parallels = same area; square from rectangle uses a semicircle, side = sqrt(l x w)

Six mistakes that lose marks

1. Rubbing out the construction arcs. The arcs are your working and your proof - leave every one visible.
2. Treating Case 3 like Case 1: if the angle is not included, always check for a second intersection, or for none.
3. Starting an ASA drawing when the two angles add to 180 degrees or more - no triangle exists.
4. Swapping incentre and circumcentre properties: bisectors point - equidistant from *sides*; perpendicular-bisector point - equidistant from *vertices*.
5. Drawing a median to a point that is not the midpoint, or an altitude that is not perpendicular. Construct the midpoint and the right angle; do not estimate them.
6. Giving only the figure with no steps of construction, or steps with no figure. The board awards both parts separately.

11. Practice - MCQs and Constructions

Multiple-choice questions (from the book's Review Exercise 17)

1. The first step in constructing an angle bisector is to: (a) draw a ray (b) label the intersections (c) measure the line (d) place the compass point on the vertex
2. A line segment joining the midpoint of one side of a triangle to its opposite vertex is called a: (a) perpendicular bisector (b) median (c) altitude (d) angle bisector
3. If the orthocentre, centroid and circumcentre of a triangle are all the same point, the triangle is: (a) scalene (b) isosceles (c) equilateral (d) right-angled
4. The centroid divides the medians in the ratio: (a) 2:1 (b) 3:1 (c) 4:1 (d) 5:1
5. The point of intersection of the angle bisectors is equidistant from the ___ of the triangle: (a) vertices (b) sides (c) altitudes (d) medians
6. The point of intersection of the perpendicular bisectors is equidistant from the ___ of the triangle: (a) altitudes (b) medians (c) sides (d) vertices
7. Altitudes of a triangle are: (a) equal in length (b) equidistant from the vertices (c) concurrent (d) perpendicular bisectors
8. If the measures of the three angles of a triangle are known, the number of triangles that can be constructed is: (a) only one (b) two (c) none (d) infinite

Construction practice

1. Construct triangle ABC with AB = 3.7 cm, BC = 2.5 cm and angle B = 60 degrees.
2. Construct triangle ABC with BC = 5.8 cm, angle A = 30 degrees and angle B = 45 degrees.
3. Construct triangle ABC (whenever possible) with angle B = 30 degrees, AB = 8 cm and AC = 4 cm.
4. Construct triangle PQR with PQ = 6 cm, QR = 4.5 cm and PR = 5.5 cm; draw its altitudes and verify their concurrency.
5. Draw rectangle PQRS with PQ = 5 cm and QR = 3.5 cm; construct a square equal in area to it.

Answer key

MCQs: 1(d), 2(b), 3(c), 4(a), 5(b), 6(d), 7(c), 8(d).

Constructions: 1) Case 1 (SAS) - angle B is included between AB and BC; one triangle. 2) Case 2 - first find angle C = 180 - 30 - 45 = 105 degrees; draw BC, build 45 degrees at B and 105 degrees at C; one triangle. 3) Case 3 - draw AB = 8 cm, build 30 degrees at B, then an arc of radius AC = 4 cm with centre A. The perpendicular distance from A to the arm is 8 x sin 30 degrees = 4 cm, exactly the radius, so the arc just touches the arm: **one** triangle, right-angled. 4) Three-sides construction first, then a perpendicular from each vertex; the altitudes meet at one point. 5) Section 9.1: side = sqrt(5 x 3.5) = sqrt(17.5), about 4.18 cm - check your measured side against this.